

A Finite Difference Approximation of Second Order Regularization of Neural-SDFs

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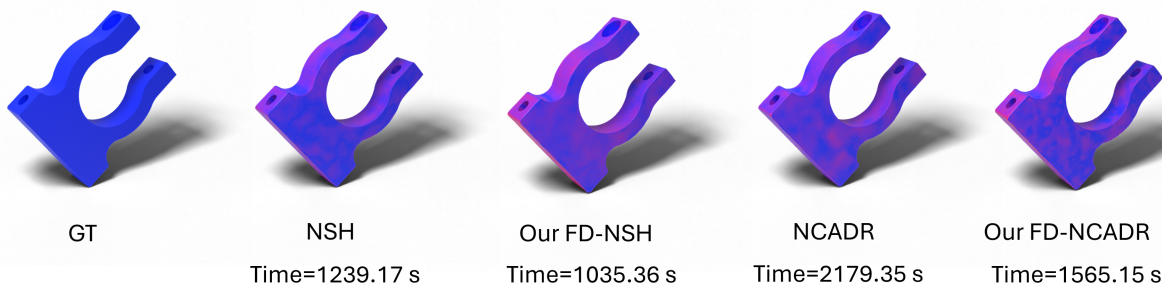


Figure 1: Neural-SDF reconstructions with Hausdorff-distance error maps (blue $\rightarrow$ low, red $\rightarrow$ high) of baseline methods NSH [2023] and NeurCADRecon [2024] compared to our proposed FD-counterparts.

Abstract

We introduce a finite-difference framework for curvature regularization in neural signed distance field (SDF) learning. Existing approaches enforce curvature priors using full Hessian information obtained via second-order automatic differentiation, which is accurate but computationally expensive. Others reduced this overhead by avoiding explicit Hessian assembly, but still required higher-order differentiation. In contrast, our method replaces these operations with lightweight finite-difference stencils that approximate second derivatives using the well known Taylor expansion with a truncation error of $O(h^2)$, and can serve as drop-in replacements for Gaussian curvature and rank-deficiency losses. Experiments demonstrate that our finite-difference variants achieve reconstruction fidelity comparable to their automatic-differentiation counterparts, while reducing GPU memory usage and training time by up to a factor of two. Additional tests on sparse, incomplete, and non-CAD data confirm that the proposed formulation is robust and general, offering an efficient and scalable alternative for curvature-aware SDF learning.

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1 Introduction

Neural signed distance fields (SDFs) are widely used for surface learning and reconstruction. However, for CAD-like objects, first-order regularizers such as the Eikonal loss are insufficient to capture sharp features and developable regions. NeurCADRecon (NCR) [Dong et al. 2024] and Neural-Singular-Hessian (NSH) [Wang et al. 2023] address this limitation using curvature-based losses derived from second-order automatic differentiation. While effective, these approaches are computationally demanding, as they require explicit evaluation of the Hessian or its determinant during training.

FlatCAD [Yin et al. 2025] proposed a more efficient alternative by formulating a curvature regularizer that avoids assembling the full Hessian. This reduces overhead compared to full Hessian methods, yet still incurs the cost of higher-order automatic differentiation.

FlatCAD also sketched an initial variant based on finite differences. We build on this idea and introduce a general finite-difference (FD) formulation for curvature regularization in neural SDFs. By combining a few forward evaluations with first-order gradients, our method enforces curvature constraints without



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requiring any second-order automatic differentiation. Recently, a rendering approach applied a similar forward-evaluation principle for higher-order estimation [Wang et al. 2024], underpinning that concept.

Our scheme is second-order accurate, framework-agnostic, and computationally significantly cheaper than auto-diff methods. Our contributions are: (1) A finite-difference discretizations of curvature regularizers for neural SDFs that achieve similar accuracy using only first-order terms. (2) An implementation that avoids higher-order auto-diff entirely, requiring only forward SDF evaluations. (3) Empirical validation on the ABC dataset showing that our method matches or improves the reconstruction quality SOTA, while reducing memory usage and training time by up to 2×.

2 Related Work

Implicit functions are a widely used representation for 3D reconstruction. Classical methods estimate signed distances from tangent planes [Hoppe et al. 1992], radial basis functions [Carr et al. 2001], or Poisson surface reconstruction [Kazhdan et al. 2006]. Neural implicit methods define surfaces as the zero level-set of a learned field, with DeepSDF [Park et al. 2019] as a seminal example. Self-supervised approaches such as IGR [Gropp et al. 2020] and SAL [Atzmon and Lipman 2019] introduced Eikonal and sign-agnostic losses, while SIREN [Sitzmann et al. 2020] improved high-frequency fitting.

Beyond first-order constraints, recent work has emphasized curvature-aware priors. DiGS [Ben-Shabat et al. 2021] penalizes divergence to promote smoothness, while other methods introduced curvature or rank-based regularizers for developability [Novello et al. 2023; Selvaraju 2024]. Neural-Singular-Hessian (NSH) [Wang et al. 2023] minimizes the smallest singular value of the Hessian to enforce rank-deficiency, and NeurCADRecon [Dong et al. 2024] uses a Gaussian curvature loss derived from the Hessian. FlatCAD [Yin and Musialski 2025; Yin et al. 2025] proposed a more efficient determinant-based regularizer that avoids explicit Hessian assembly but still relies on higher-order derivatives. These methods improve fidelity and feature preservation but suffer from the computational burden of second-order derivatives.

3 Our Approach

We propose a finite-difference regularization framework for learning neural signed distance fields (SDFs) from unoriented point clouds. Following the standard self-supervised setup for neural implicit reconstruction, we combine a point-based Dirichlet condition and an Eikonal loss to ensure surface fidelity and valid SDF behavior. Our core contribution is a second-order regularizer based on finite differences. Instead of computing Hessians via costly second-order automatic differentiation, we approximate directional second derivatives from local SDF evaluations using Taylor expansion stencils. This enables lightweight reformulations of curvature-based penalties such as those in NRC [Dong et al. 2024] and NSH [Wang et al. 2023], while achieving $O(h^2)$ accuracy and avoiding full second-order computation graphs.

3.1 Differential-Geometric Background

A surface $\mathcal{M} \subset \mathbb{R}^3$ is represented as the zero-level set of a signed distance field (SDF):

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad \mathcal{M} = \{\mathbf{x} \mid f(\mathbf{x}) = 0\}.$$

A valid SDF satisfies the Eikonal condition $\|\nabla f\| = 1$ in a neighborhood around $\mathcal{M}$, implying zero second derivative along the normal:

$$\partial_{nn}f = \mathbf{n}^\top H_f \mathbf{n} = 0, \quad \mathbf{n} = \frac{\nabla f}{\|\nabla f\|},$$

where H_f denotes the Hessian. Projecting H_f onto an orthonormal tangent basis $(\mathbf{u}, \mathbf{v})$ yields the shape operator

$$S_{\mathcal{M}} = \begin{pmatrix} \mathbf{u}^\top H_f \mathbf{u} & \mathbf{u}^\top H_f \mathbf{v} \\ \mathbf{v}^\top H_f \mathbf{u} & \mathbf{v}^\top H_f \mathbf{v} \end{pmatrix},$$

whose eigenvalues are the principal curvatures κ_1, κ_2 with Gaussian curvature $K = \kappa_1 \kappa_2 = \det(S_{\mathcal{M}})$. Any uniformly random orthogonal completion $\mathbf{u}, \mathbf{v}$ of the gradient normal is equally valid, and expectation over such random frames recovers the correct curvature measure [Yin et al. 2025].

To avoid second-order automatic differentiation, we approximate the entries of S with finite-difference stencils. For step size $h > 0$ and tangent directions $\mathbf{u}, \mathbf{v}$, the second derivatives are approximated as

$$f_{uu} \approx \frac{f(\mathbf{x}_0 + h\mathbf{u}) - 2f(\mathbf{x}_0) + f(\mathbf{x}_0 - h\mathbf{u})}{h^2},$$

$$f_{vv} \approx \frac{f(\mathbf{x}_0 + h\mathbf{v}) - 2f(\mathbf{x}_0) + f(\mathbf{x}_0 - h\mathbf{v})}{h^2},$$

$$f_{uv} \approx \frac{1}{4h^2} \left(f(\mathbf{x}_0 + h\mathbf{u} + h\mathbf{v}) - f(\mathbf{x}_0 + h\mathbf{u} - h\mathbf{v}) - f(\mathbf{x}_0 - h\mathbf{u} + h\mathbf{v}) + f(\mathbf{x}_0 - h\mathbf{u} - h\mathbf{v}) \right).$$

By Taylor expansion, these approximations incur an error of $O(h^2)$.

3.2 FD-NeurCADRecon

NeurCADRecon (NRC) [Dong et al. 2024] enforces piecewise-developability by penalizing the Gaussian curvature of the learned SDF. In the original formulation, this is expressed via the Hessian H_f as

$$K = - \frac{\det \begin{pmatrix} H_f(\mathbf{x}_0) & \nabla f(\mathbf{x}_0)^\top \\ \nabla f(\mathbf{x}_0) & 0 \end{pmatrix}}{\|\nabla f(\mathbf{x}_0)\|^4}.$$

Evaluating this term requires explicit Hessian-vector products through second-order automatic differentiation, which is costly in both runtime and memory.

In our finite-difference (FD) formulation, we avoid second-order derivatives entirely. We first construct a local tangent frame $(\mathbf{u}, \mathbf{v})$ at an off-surface shell point $\mathbf{x}_0$, and approximate the directional second derivatives f_{uu}, f_{vv}, f_{uv} via central-difference stencils of nearby function evaluations. Gaussian is then estimated as

$$K_{\text{FD}}(\mathbf{x}_0) = \frac{f_{uu}f_{vv} - f_{uv}^2}{\|\nabla f(\mathbf{x}_0)\|^4},$$

which is second-order accurate by Taylor expansion with truncation error $O(h^2)$. This replacement yields the same geometric bias as the original NRC loss while reducing both memory consumption and wall-clock training time.

3.3 FD-Neural-Singular-Hessian

The Neural-Singular-Hessian (NSH) regularizer [Wang et al. 2023] promotes developability by minimizing $\det(H_f)$ near the surface, thereby forcing the Hessian to be rank-deficient. This suppresses spurious curvature and aligns normals in a piecewise-developable manner. The drawback is that explicit determinant evaluation requires full Hessian computation.

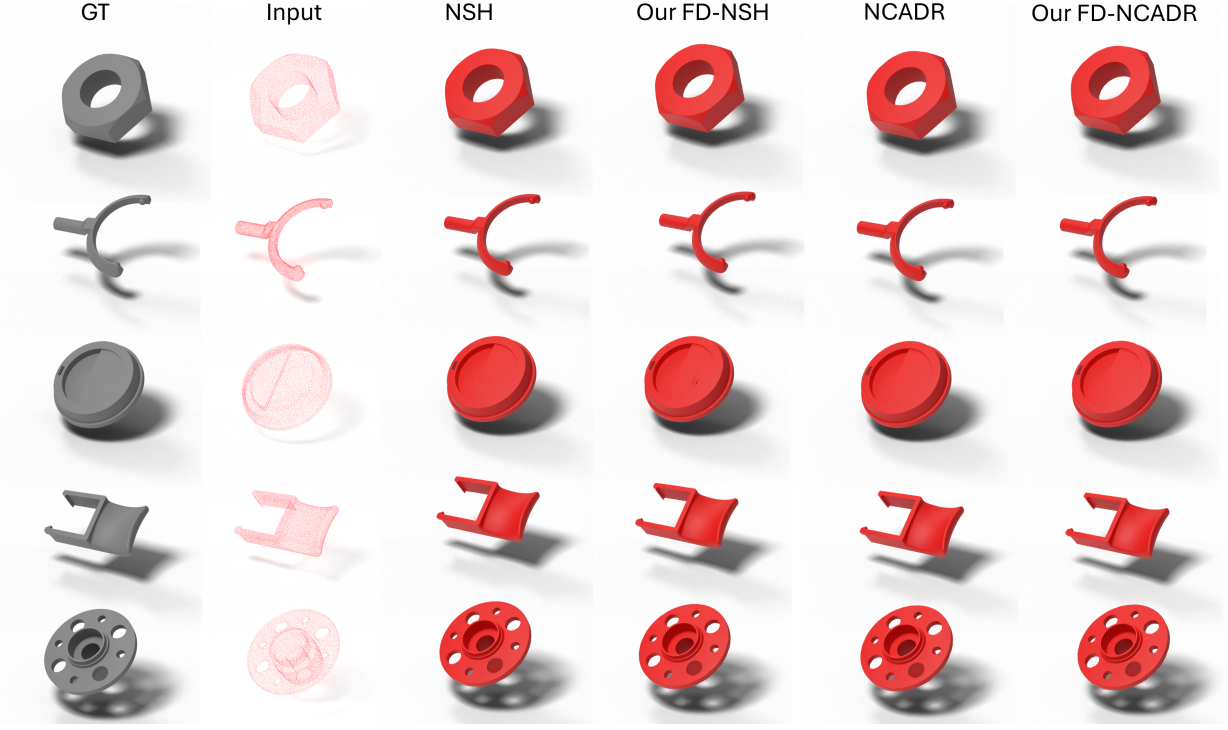


Figure 2: Comparison with state-of-the-art surface reconstruction. FD and AD proxies yield comparable accuracy. Our FD approach avoids second-order differentiation while maintaining efficiency.

We reformulate this regularizer in the tangent frame $(\mathbf{u}, \mathbf{v})$ using finite differences. Specifically, the projected determinant is approximated as

$$D_{\text{FD}}(\mathbf{x}_0) = f_{uu}f_{vv} - f_{uv}^2,$$

where f_{uu}, f_{vv}, f_{uv} are obtained via the same central-difference stencils as above. By construction,

$$D_{\text{FD}}(\mathbf{x}_0) = (\mathbf{u}^\top H_f \mathbf{u})(\mathbf{v}^\top H_f \mathbf{v}) - (\mathbf{u}^\top H_f \mathbf{v})^2 + O(h^2),$$

guaranteeing second-order consistency. As D_{FD} approaches zero, at least one principal curvature vanishes, ensuring zero normal curvature while preserving sharp features. This finite-difference formulation thus retains the geometric effect of NSH but removes the overhead of second-order automatic differentiation.

3.4 Practical Loss Implementation

In practice, we leverage the property $\|\nabla f\| \approx 1$ near the surface, simplifying denominators in curvature terms. We found that early training gradients introduce oscillations, and in FD-NRC we replace $\|\nabla f(\mathbf{x}_0)\|$ by 1 for near-surface points for stability.

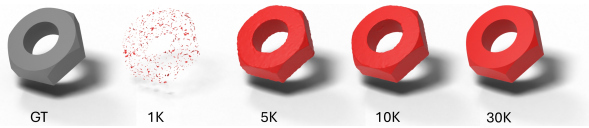


Figure 3: Qualitative reconstructed results on point clouds with varying levels of sparsity.

Our total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DM}} + \lambda_{\text{DNM}} \mathcal{L}_{\text{DNM}} + \lambda_{\text{eik}} \mathcal{L}_{\text{eik}} + \lambda_{\text{fd}} \mathcal{L}_{\text{fd}}, \quad (1)$$

where $\mathcal{L}_{\text{DM}}$ is a Dirichlet condition anchoring input points to the zero-level set, $\mathcal{L}_{\text{DNM}}$ a non-manifold penalty following [Atzmon and Lipman 2020], $\mathcal{L}_{\text{eik}}$ the Eikonal constraint [Gropp et al. 2020], and $\mathcal{L}_{\text{fd}}$ our finite-difference curvature regularizer.

4 Experiments and Results

We validate our finite-difference regularizers FD-NCR and FD-NSH against their automatic-differentiation counterparts NeurCADRecon (NRC) [2024] and Neural-Singular Hessian (NSH) [2023].

Datasets. Experiments are performed on two subsets of the ABC dataset [Koch et al. 2019] to assess accuracy, robustness, and efficiency. These subsets of 100 shapes each are: a pseudo-random *1 MB set*, and a curated *5 MB set* (3.5–9.5 MB average) with clean topology and sharp features. For each mesh, we sample 30k surface points; 20k are used per iteration, with 20k off-surface samples drawn uniformly in the bounding box.

Methods and Setup. All methods are trained in a unified framework with a 4-layer SIREN MLP [Sitzmann et al. 2020] (256 hidden units), Adam optimizer (5×10^{-5}), and early stopping if Chamfer Distance does not improve for 1500 iterations. Hyperparameters follow the originals.

Runtime results on the 1 MB set are measured on an NVIDIA H100 GPU. Qualitative and ablation studies additionally use A100, L4, and T4 GPUs; only H100 numbers are reported for consistency.

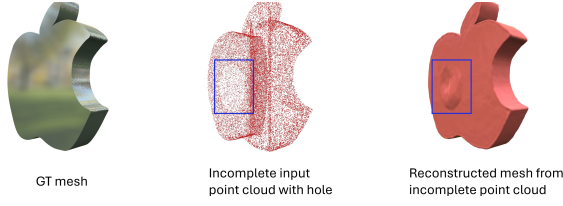


Figure 4: Reconstruction from incomplete input clouds. Missing planar regions are plausibly filled while preserving global continuity.

We report Chamfer Distance (CD, $\times 10^3$, lower is better), F1 Score (F1, $\times 10^2$, higher is better), and Normal Consistency (NC, $\times 10^2$, higher is better), averaged over 100 shapes per subset. On the 1 MB set we also measure GPU memory, per-iteration time, and convergence time (best iteration $\times$ iteration time).

Quantitative and Qualitative. FD-NCR and FD-NSH achieve accuracy on par with their AD versions, while reducing memory and runtime (Table 1 and supplement). Hausdorff heatmaps (Fig. 1) show reduced local errors, and reconstructions (Fig. 2) confirm comparable fidelity to NRC and NSH.

With reduced sampling (30k $\rightarrow$ 10k, 5k, 1k points), our method maintains quality at 10k and 5k; only 1k shows strong degradation (Fig. 3, Table: supplement). With holes excised from the input, geometry degrades gracefully: CD increases by $\sim 64\%$ and NC drops by 0.7%, while global topology is preserved (Fig. 4, Table: supplement). On non-CAD data (e.g., Armadillo), FD-NCR reconstructs smooth, topologically correct surfaces comparable to NRC but at about half the runtime (cf. supplement for figures and table).

5 Conclusion

We presented a finite-difference framework for curvature regularization of neural signed distance fields. By approximating second derivatives with FD-stencils, our method avoids higher-order automatic differentiation while retaining second-order accuracy. We demonstrated results on two SOTA methods that achieve comparable or improved reconstruction quality with substantially lower memory and runtime cost. Future work includes extending the framework to adaptive sampling strategies and exploring applications beyond surface reconstruction. Please refer to supplemental material for more quantitative results.

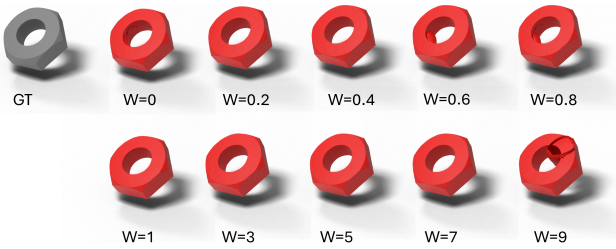


Figure 5: Reconstructions with different regularization weights. Results remain stable across a wide range.

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Table 1: Accuracy and efficiency (full table in supplemental).

Method	NC $\uparrow$	CD $\downarrow$	F1 $\uparrow$	Conv. [s]	Mem. [GB]
NSH [Wang et al. 2023]	93.93	2.74	91.91	559.41	6.09
NSH-FD (Our)	94.96	2.93	89.90	362.95	4.30
NCR [Dong et al. 2024]	93.71	2.65	90.63	391.23	6.06
NCR-FD (Our)	93.41	4.10	84.78	331.29	4.03